Korba Coalfield

🏭 Introduction

The **Korba Coalfield** is one of the most important coal-producing regions in India, located in Chhattisgarh. It is primarily operated by South Eastern Coalfields Limited, a subsidiary of Coal India Limited.

The coalfield is known for its **large reserves of non-coking (thermal) coal**, which are extensively used for electricity generation. It hosts some of India's largest opencast mines, including **Gevra, Dipka, and Kusmunda**, making it a major contributor to the country's coal production. Due to its high output and proximity to power plants, Korba is often referred to as the **"Power Capital of India."**

🌍 Geographical Overview – Korba Coalfield

The Korba Coalfield is located in the **Korba district** of Chhattisgarh and forms part of the **Hasdeo–Arand basin**, one of the richest coal-bearing regions in India.

📍 Location & Setting

- Situated in central-eastern India within the Korba region
- Well connected by rail (Gevra Road, Korba) and road networks
- Close to major industrial and power generation hubs

🌄 Physical & Geological Features

- Lies in a **plateau region with gently undulating terrain**
- Contains **thick and extensive coal seams** suitable for opencast mining
- Coal type: **non-coking thermal coal**

🌊 Nearby Natural Features

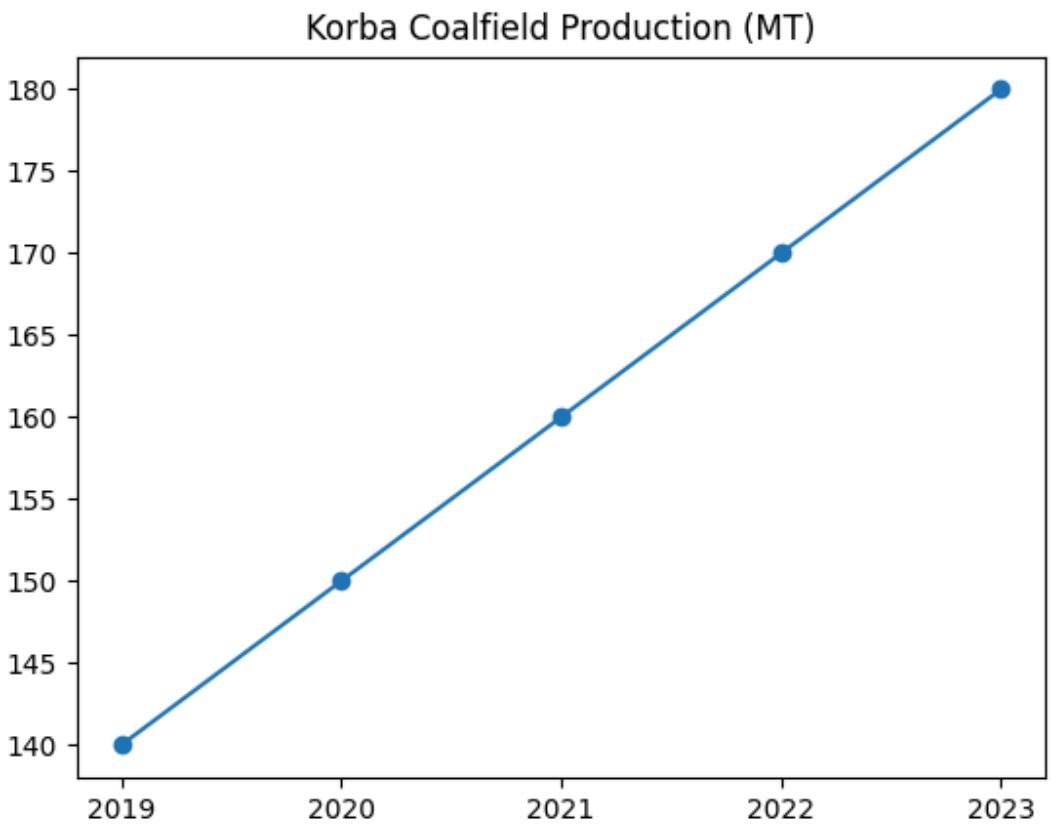
- Located near the Hasdeo River, a tributary of the Mahanadi River
- Surrounded by major thermal power plants and industrial areas

🌦️ Climate

- Tropical climate
- Hot summers (up to ~47°C)
- Moderate to heavy monsoon rainfall

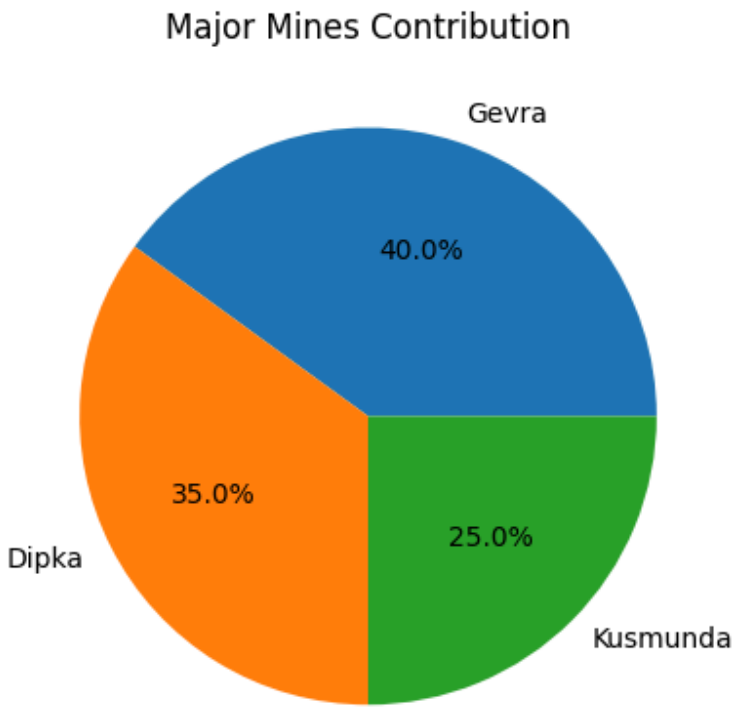
- Mild winters

📊 1. Production Trend (MT)



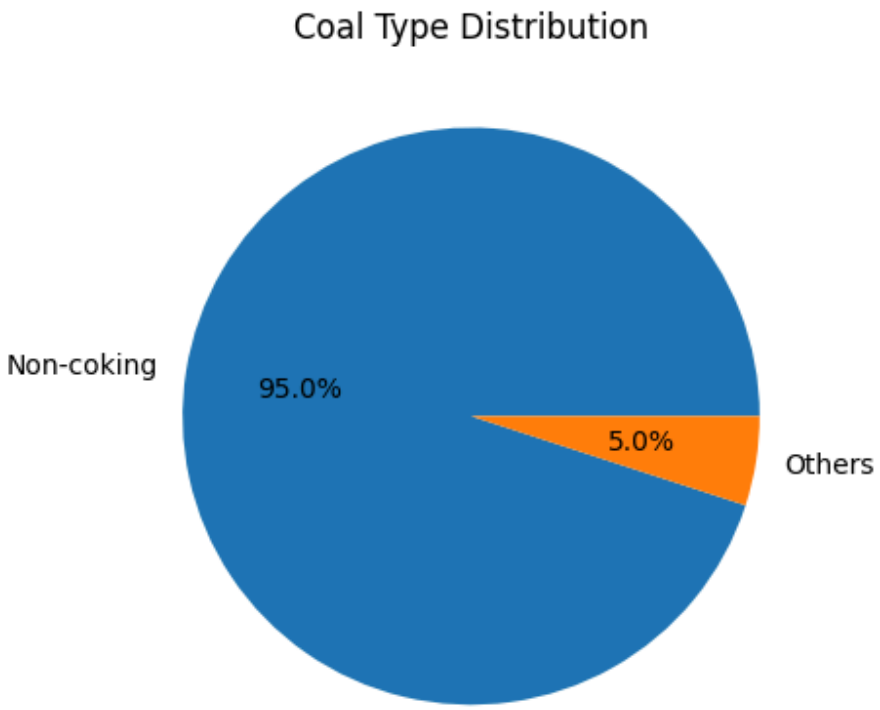
This chart shows the overall coal production of Korba coalfield under South Eastern Coalfields Limited.
The increasing trend reflects expansion of major mines like Gevra and Kusmunda.
Significance: Useful for understanding regional production growth and energy supply trends.

📊 2. Major Mines Contribution



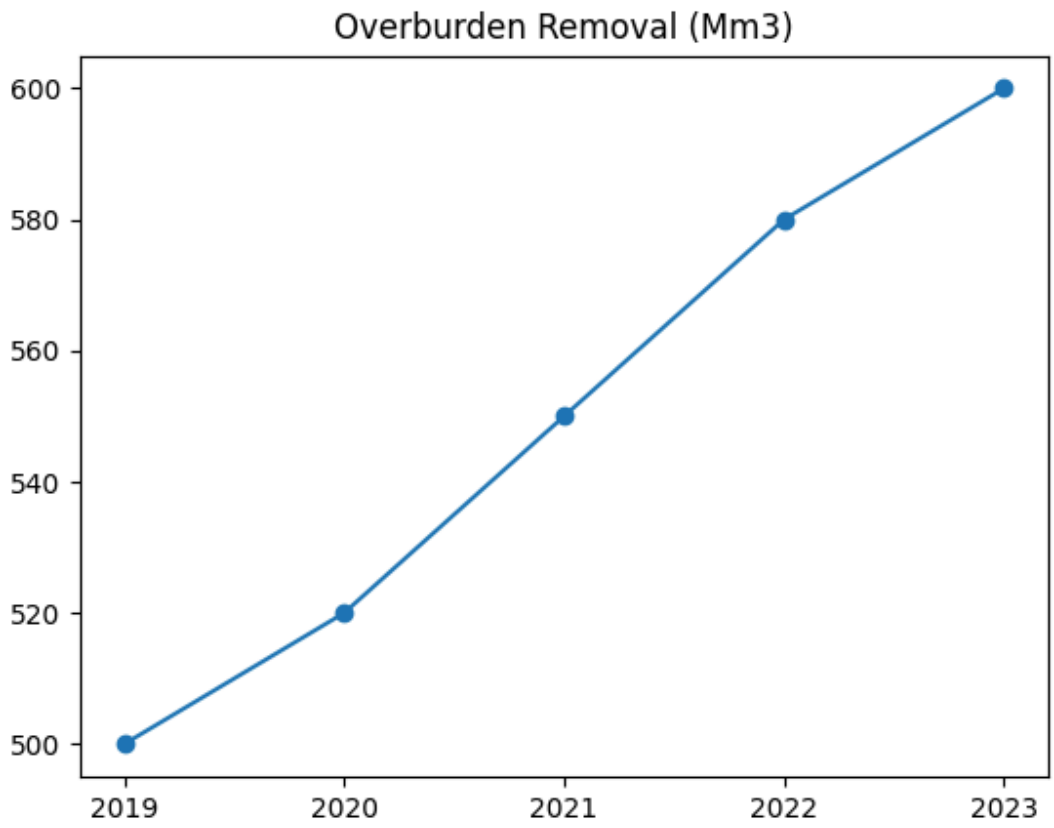
Represents the share of major mines such as Gevra, Dipka, and Kusmunda.
Gevra contributes the largest portion due to its massive capacity.
Significance: Helps identify key production centers within the coalfield.

📊 3. Coal Type Distribution



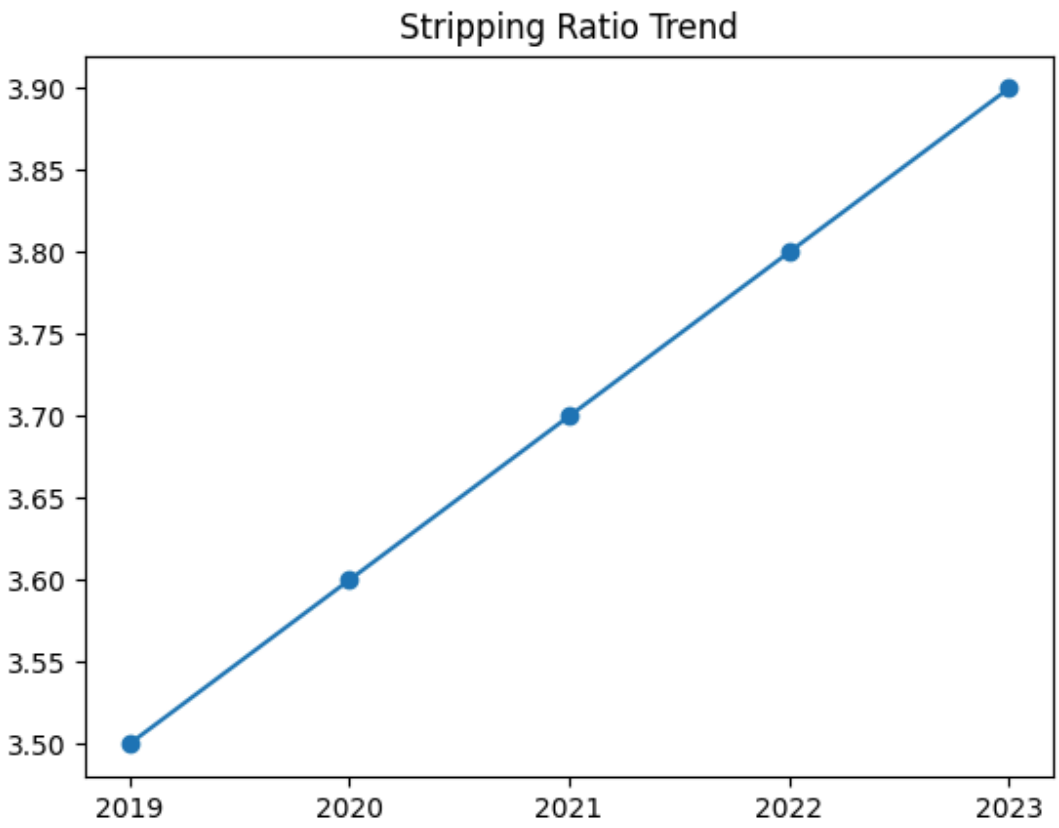
Shows dominance of non-coking (thermal) coal in Korba.
This coal is mainly used for power generation.
Significance: Determines end-use sector and highlights importance in electricity production.

📊 4. Overburden Removal (Mm³)



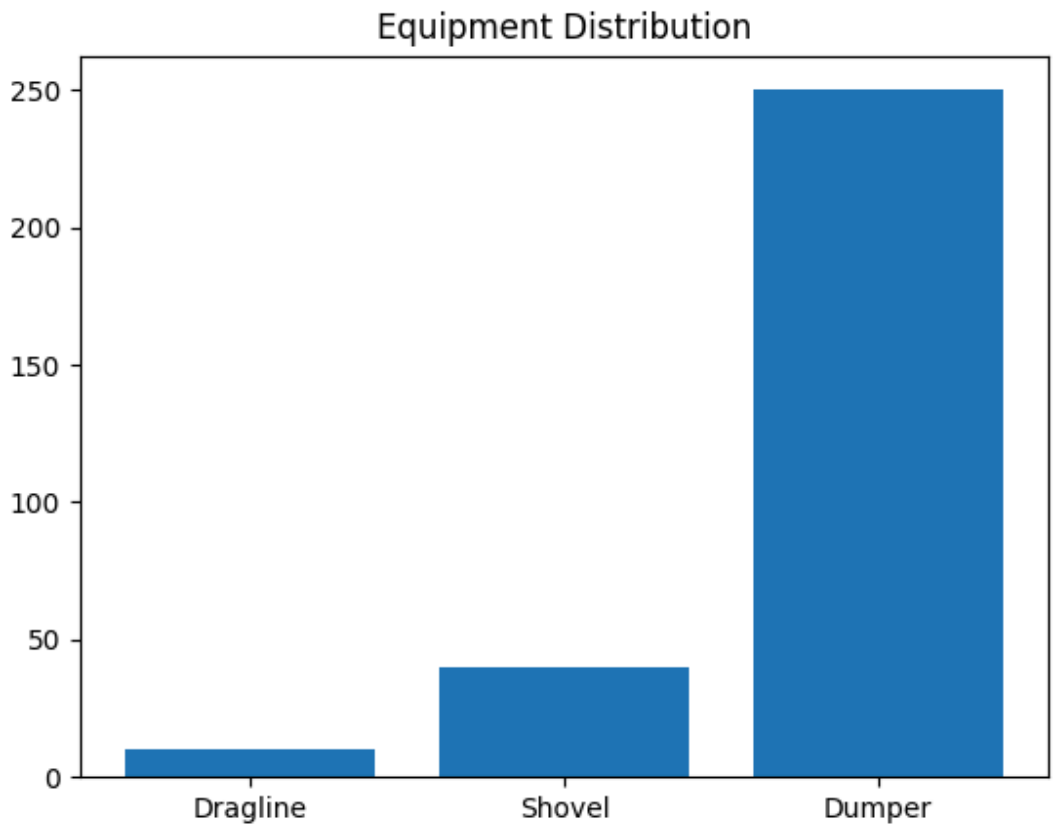
Displays the large volume of overburden removed annually.
Reflects the large-scale opencast nature of mining in Korba.
Significance: Important for dump design, equipment planning, and cost estimation.

📊 5. Stripping Ratio Trend



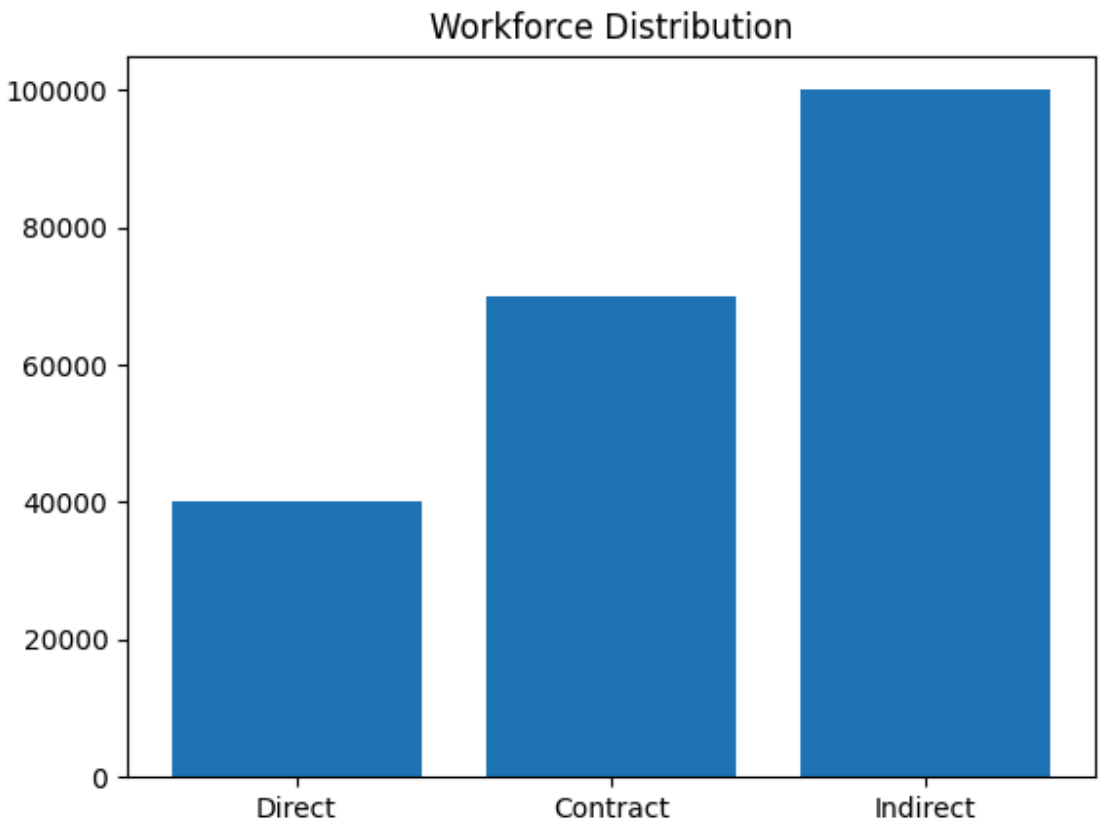
Shows gradual increase in stripping ratio over time.
Indicates deeper mining and increasing operational cost.
Significance: Key parameter for economic feasibility and mine planning.

📊 6. Equipment Distribution



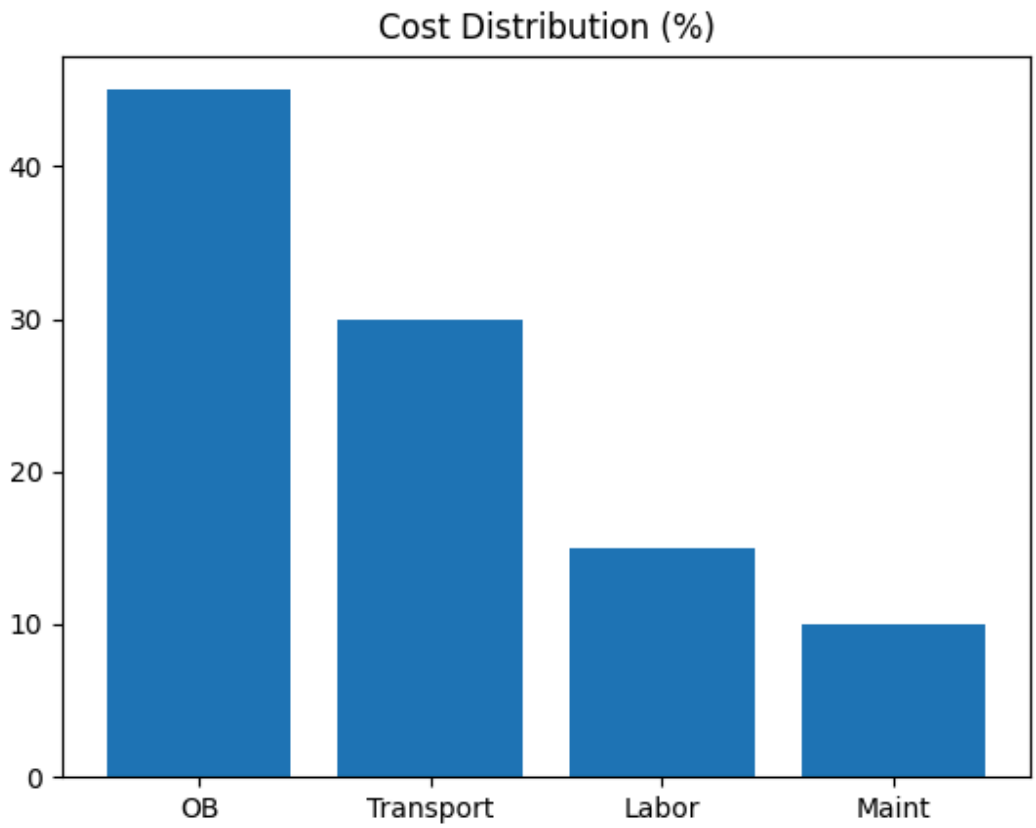
Represents deployment of heavy machinery like draglines, shovels, and dumpers.
Dumpers dominate due to massive material movement.
Significance: Useful for studying large-scale mechanized mining systems.

📊 7. Workforce Distribution



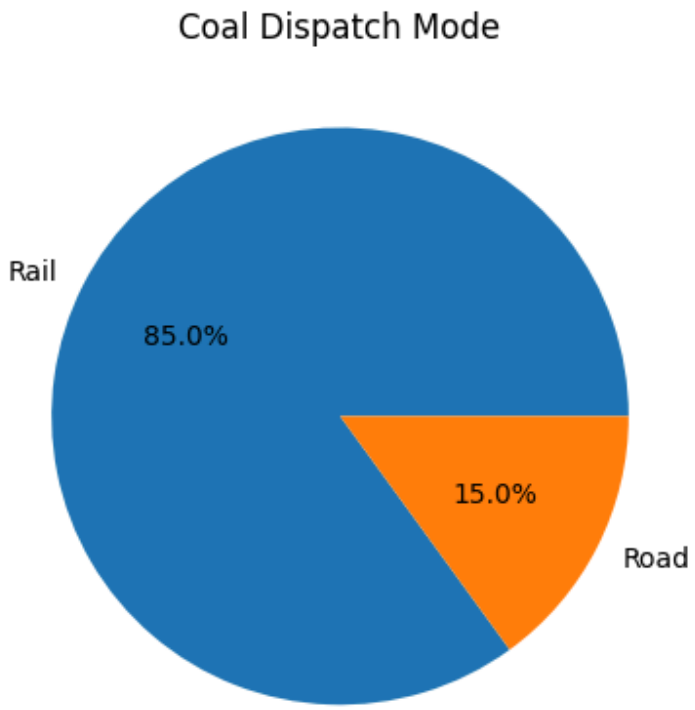
Displays manpower distribution across different employment categories.
Highlights large employment generated by mining activities.
Significance: Important for manpower planning and socio-economic analysis.

📊 8. Cost Distribution (%)



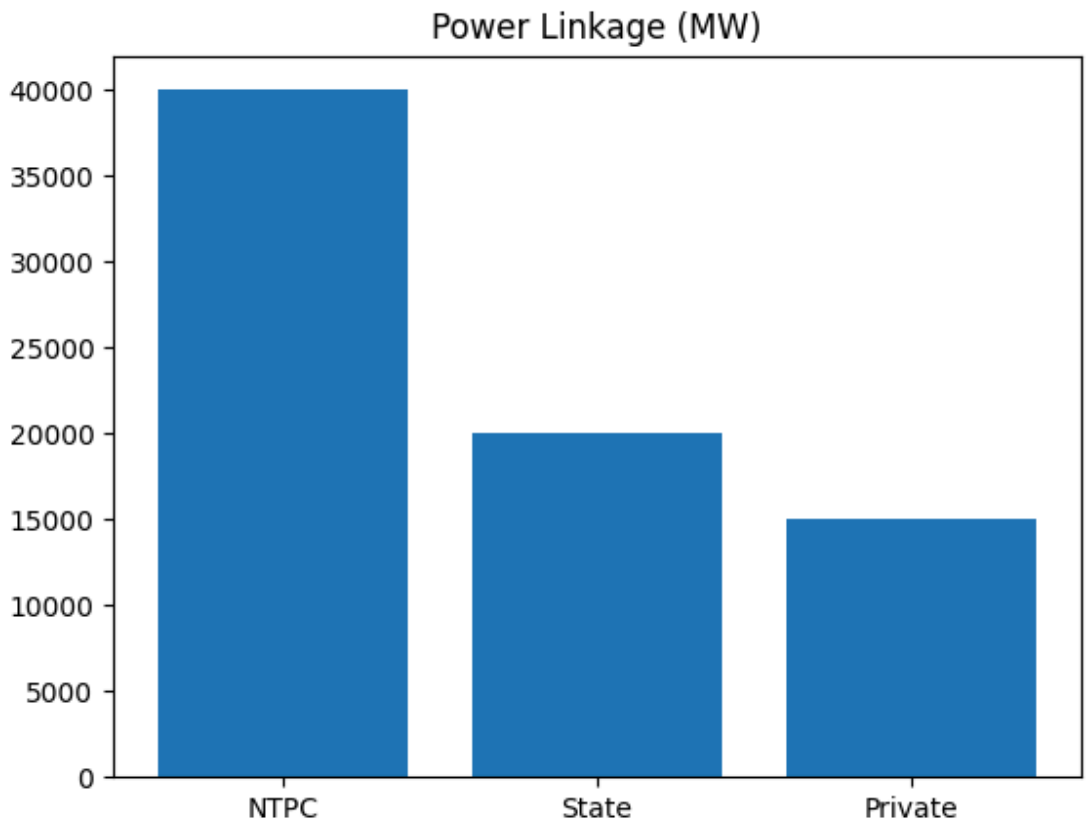
Shows breakdown of mining costs into major components.
Overburden removal forms the largest share.
Significance: Helps in cost optimization and improving operational efficiency.

📊 9. Coal Dispatch Mode



Represents transportation of coal via rail and road.
Rail transport dominates due to bulk handling efficiency.
Significance: Important for logistics planning and reducing transportation cost.

📊 10. Power Linkage (MW)



Shows coal supply linkage to various power sectors.
Korba coalfield supports multiple large thermal power plants.
Significance: Highlights its critical role in India's energy infrastructure.